Sri Sivasubramaniya Nadar College of Engineering, Chennai

(An Autonomous Institution affiliated to Anna University)

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Experiment 7

Bagging, Boosting, and Stacked Ensemble Models

1. Aim and Objective

- To understand ensemble learning strategies: **Bagging**, **Boosting**, and **Stacking**.
- To implement a **Bagging classifier** using a Decision Tree as the base estimator.
- To implement **Boosting classifiers** – AdaBoost and Gradient Boosting.
- To build a **Stacked Ensemble model** using multiple heterogeneous base learners.
- To compare ensemble models in terms of accuracy, stability, and generalization using 5-Fold Cross-Validation.
- To analyze the effect of ensemble methods on bias and variance.

2. Dataset Description

Wisconsin Diagnostic Breast Cancer Dataset

- Total samples: 569
- Features: 30 numerical attributes
- Target classes: Malignant (M = 1) and Benign (B = 0)
- Class distribution: 357 Benign, 212 Malignant

Dataset Link: https://archive.ics.uci.edu

3. Preprocessing Steps

- The dataset was loaded from `wdbc.data` with 569 instances and 32 attributes (id, diagnosis, and 30 morphological features).
- Feature column names were manually assigned including `id`, `diagnosis`, and 30 numerical morphological features (mean, standard error, and worst values).
- The target variable `diagnosis` was mapped from categorical strings (`M` $\rightarrow$ 1, `B` $\rightarrow$ 0).
- Non-predictive columns (`id`) were dropped, yielding a feature matrix X of shape $(569, 30)$.
- A correlation heatmap was plotted to visualize inter-feature relationships.
- The dataset was split into training and testing sets using an 80–20 ratio with `random_state=42` and stratified sampling to preserve class proportions.

4. Implementation Details

- **Bagging:** `BaggingClassifier` with a `DecisionTreeClassifier` base estimator. Hyperparameter tuning was performed using `GridSearchCV` over `n_estimators` $\in \{10, 50, 100\}$, `max_samples` $\in \{0.5, 0.7, 1.0\}$, `max_features` $\in \{0.5, 0.7, 1.0\}$, and `bootstrap` $\in \{\text{True}, \text{False}\}$.
- **AdaBoost:** `AdaBoostClassifier` with a depth-1 Decision Tree (stump) as the weak learner. Tuned over `n_estimators` $\in \{50, 100, 200\}$ and `learning_rate` $\in \{0.01, 0.1, 1.0\}$.
- **Gradient Boosting:** `GradientBoostingClassifier` tuned over `n_estimators` $\in \{100, 200\}$, `learning_rate` $\in \{0.01, 0.1\}$, `max_depth` $\in \{3, 5\}$, and `subsample` $\in \{0.8, 1.0\}$.
- **Stacking:** `StackingClassifier` with base models Decision Tree and KNN, and `LogisticRegression` as the meta-learner. The meta-learner's regularization parameter $C \in \{0.01, 0.1, 1, 10\}$ and `solver` $\in \{\texttt{lbfgs}, \texttt{liblinear}\}$ were tuned.
- All models were evaluated using 5-Fold Cross-Validation (`cv=5`, `scoring='accuracy'`) and subsequently assessed on the held-out test set.

5. Visualizations

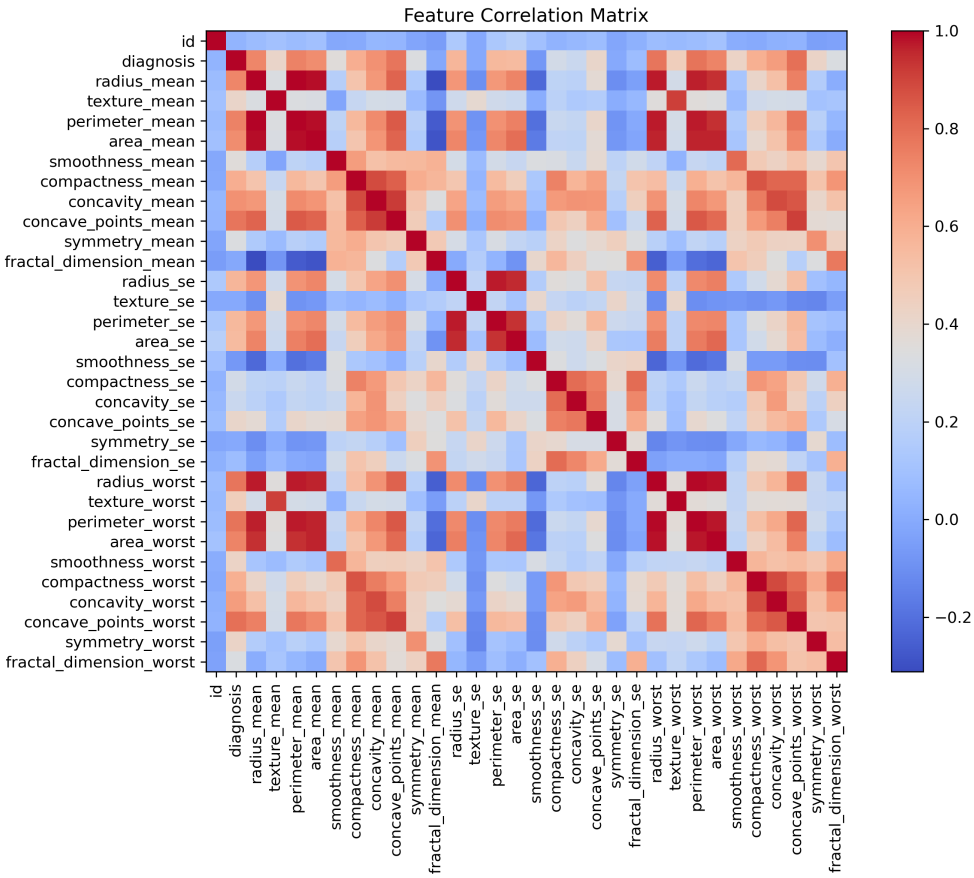
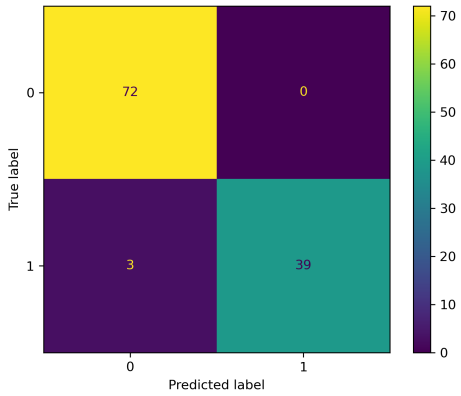
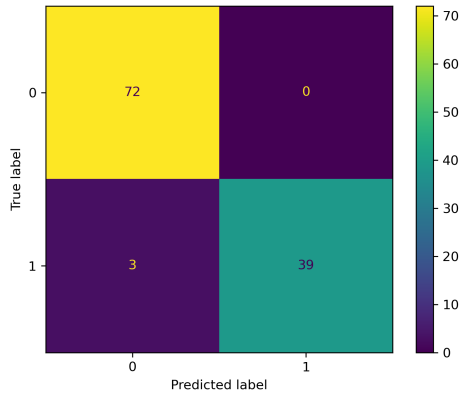


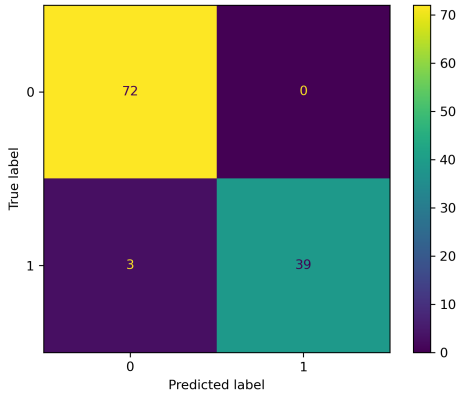
Figure 1: Feature Correlation Matrix



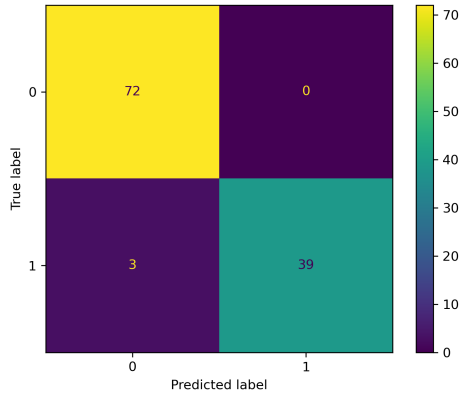
(a) Confusion Matrix – Bagging



(b) Confusion Matrix – AdaBoost



(c) Confusion Matrix – Gradient Boosting



(d) Confusion Matrix – Stacking

Figure 2: Confusion Matrices for All Ensemble Models

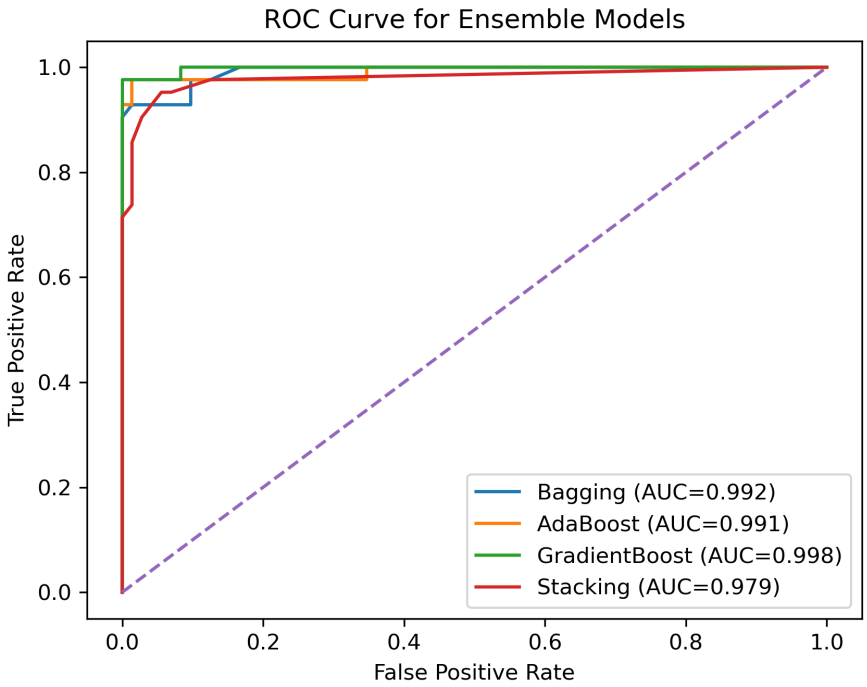


Figure 3: ROC Curves for All Ensemble Models

6. Hyperparameter Tuning Results

Bagging Results

Table 1: Bagging Hyperparameter Evaluation using 5-Fold Cross-Validation

`n_estimators`	`max_samples`	Avg CV Accuracy (%)	Avg CV F1 Score
50	0.7	96.92	0.9687
100	0.7	96.92	0.9687
50	1.0	96.92	0.9687
50	0.7	96.48	0.9643
100	0.5	96.26	0.9621

Boosting Results

Table 2: Boosting Hyperparameter Evaluation using 5-Fold Cross-Validation

`n_estimators`	`learning_rate`	Avg CV Accuracy (%)	Avg CV F1 Score
AdaBoost			
200	1.0	96.48	0.9643
50	1.0	96.26	0.9620
100	1.0	96.04	0.9598
200	0.1	95.16	0.9509
100	0.1	95.16	0.9509
Gradient Boosting (`max_depth=5, subsample=0.8`)			
200	0.1	96.92	0.9687
200	0.1	96.26	0.9620
100	0.1	96.26	0.9620
200	0.1	95.60	0.9554

Stacked Ensemble Results

Table 3: Stacked Ensemble Hyperparameter Evaluation using 5-Fold Cross-Validation

Base Models	Meta Learner	Avg CV Accuracy (%)	Avg CV F1 Score
DT + KNN	LR ($C = 10$, liblinear)	93.41	0.9334
DT + KNN	LR ($C = 0.1$, lbfgs)	93.41	0.9334
DT + KNN	LR ($C = 1$, lbfgs)	93.19	0.9311
DT + KNN	LR ($C = 0.01$, liblinear)	93.19	0.9311

7. Performance Comparison

Table 4: Performance Comparison of Ensemble Models on Test Set

Model	Accuracy (%)	Precision	Recall	F1 Score
Bagging (Tuned)	97.37	0.97	0.97	0.97
AdaBoost (Tuned)	97.37	0.97	0.97	0.97
Gradient Boosting (Tuned)	96.49	0.97	0.96	0.96
Stacking (Tuned)	93.86	0.94	0.94	0.94

8. 5-Fold Cross-Validation Performance Comparison

Table 5: 5-Fold Cross-Validation Accuracy Comparison (Best Models)

Model	Best CV Accuracy (%)	Test Accuracy (%)
Bagging	96.92	97.37
AdaBoost	96.48	97.37
Gradient Boosting	96.92	96.49
Stacking	93.41	93.86

9. Observation Questions

How does Bagging reduce variance?

Bagging (Bootstrap Aggregating) reduces variance by training multiple Decision Trees on different bootstrap samples of the training data. Since each tree sees a slightly different dataset, their errors are less correlated. Aggregating their predictions (majority voting) averages out these errors, resulting in a more stable and accurate model. In this experiment, Bagging achieved 97.37% accuracy, outperforming a single Decision Tree (around 94.7%), demonstrating effective variance reduction.

How does Boosting address model bias?

Boosting trains models sequentially, where each new learner focuses on correcting the errors of the previous one. Methods like AdaBoost assign higher weights to misclassified samples, while Gradient Boosting fits residual errors. This iterative process reduces bias and improves model performance. In this experiment, AdaBoost with weak learners achieved 97.37% accuracy after multiple rounds, showing how boosting converts weak learners into a strong model.

Why does stacking benefit from heterogeneous models?

Stacking combines different types of models to capture diverse patterns in the data. Since each model makes different errors, a meta-learner can learn how to best combine their predictions. This improves overall performance. However, in this experiment, stacking achieved 93.86% accuracy, suggesting limited diversity among base models or insufficient tuning of the meta-learner.

Which ensemble method performed best and why?

Bagging and AdaBoost performed best with 97.37% accuracy and similar evaluation metrics. Bagging reduced variance through averaging multiple trees, while AdaBoost reduced bias by focusing on difficult samples. Gradient Boosting achieved slightly lower accuracy (96.49%), possibly due to mild underfitting. Stacking performed the lowest (93.86%), likely due to limited model diversity and dataset size.

10. Conclusion

Bagging, Boosting, and Stacking were implemented on the Wisconsin Breast Cancer dataset and evaluated using cross-validation and test data. Bagging and AdaBoost achieved the highest accu-

racy of 97.37%, with Gradient Boosting close at 96.49%. Bagging reduced variance effectively, while Boosting reduced bias through iterative learning. Stacking showed lower performance (93.86%), indicating the need for better model diversity and tuning. Overall, ensemble methods outperformed single models, and the choice depends on whether bias or variance is the dominant issue.